# Worksheet 2A: Lines, Coordinates & Vectors — Solutions

AIML Foundations Mathematics
Dublin and Dún Laoghaire ETB
Instructor: Josh Aaron

*Instructor Copy — Not for Distribution*

**Note to instructor:** Answers below reflect the expected responses. Where a question asks for reasoning or explanation, sample responses are given. Accept any mathematically equivalent form unless the question specifies a particular form.

## Part A

1. (Graphing exercise)
2. (a) IV (b) III (c) y-axis (d) II (e) x-axis
3. (a) 5 (b) 5 (c) $\sqrt{52} = 2\sqrt{13} \approx 7.21$ (d) 10
4. (a) (4, 5) (b) (1, 1) (c) (-2, 3)
5. (6, -2)
6. (-1, -2)

## Part B

7. (a) (4, 2) (b) (2, 6) (c) (-2, -6) (d) (6, 8) (e) (-3, 6) (f) (9, 2)
8. (a) 5 (b) 13 (c) $2\sqrt{2} \approx 2.83$ (d) $\sqrt{2} \approx 1.41$
9. (a) (3/5, 4/5) = (0.6, 0.8) (b) (0, 1)
10. (a) (1, 7) (b) Both equal 5 (c) $\sqrt{50} = 5\sqrt{2} \approx 7.07$ (d) No; triangle inequality
11. (4, 7)
12. (a) $\mathbf{v}$ (b) $\mathbf{0}$ (c) $-\mathbf{v}$
13. (3, 4)
14. k = 6

## Part C

15. (a) 2 (b) 2 (c) -1 (d) 0 (e) undefined
16. (a) positive (b) negative (c) zero (d) undefined
17. (a) y = -2x + 7 (b) y = (3/2)x - 6 (c) y = (-1/4)x + 2 (d) y = 5x + 3
18. (a) y = 3x - 2 (b) y = $-\frac{1}{2}$x + 4 (c) y = 5

19. (a) $y = 2x + 3$ (b) $y = -x + 7$ (c) $y = \frac{3}{4}x - 4$ (d) $y = 6$
20. (a) $y = 3x - 1$ (b) $y = -x + 5$ (c) $y = 2x + 1$
21. (a) x-int: 3, y-int: -6 (b) x-int: 4, y-int: 3 (c) x-int: 6, y-int: 3
22. (a) Both 3 (b) Parallel (c) No
23. $-\frac{1}{2}$
24. (a) $y = 2x - 1$ (b) $y = -\frac{1}{3}x + 3$

# Part D

25. (2, 5)
26. (6, 4)
27. (p, q) = (3.2, 2.2) or (16/5, 11/5)
28. Infinitely many solutions (same line)
29. No solution (parallel lines)
30. (t, u) = (4.67, 2.67) or (14/3, 8/3)
31. Coffee: €2.60, Tea: €2.20
32. Length: 9 cm, Width: 5 cm
33. Approximately (2, 3)
34. (a) Lines intersect at one point (b) Lines are parallel (c) Lines are identical
35. Same slope (2), different y-intercepts — parallel lines never meet
36. (a) Example: $y = x + 3$ and $y = 2x + 1$ (b) Example: $y = 2x + 1$ and $y = 2x + 5$ (c) Example: $y = 2x + 1$ and $2y = 4x + 2$

—